EXPERIMENT 2: MONOALPHABETIC SUBSTITUTION CIPHER

1. Aim

To write and execute a C program to implement the monoalphabetic substitution cipher, in which each plaintext letter is replaced by a unique ciphertext letter.

2. Algorithm

1. Start the program.
2. Initialize a substitution alphabet containing 26 unique letters.
3. Read the plaintext.
4. Replace each plaintext letter with its corresponding letter in the substitution alphabet.
5. Display the encrypted text.
6. Stop the program.

CODE:

```
import subprocess
code = r'''
#include <stdio.h>

int main()
{
    char text[] = "HELLO";
    char key[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
    int i, index;

    printf("Plaintext: %s\n", text);
    printf("Substitution alphabet: %s\n", key);

    for (i = 0; text[i] != '\0'; i++)
    {
        if (text[i] >= 'A' && text[i] <= 'Z')
        {
            index = text[i] - 'A';
            text[i] = key[index];
        }
    }
  printf("Ciphertext: %s\n", text);

    return 0;
}
'''

# Save the C program
with open("monoalphabetic.c", "w") as f:
    f.write(code)

# Compile the C program
compile_result = subprocess.run(
    ["gcc", "monoalphabetic.c", "-o", "monoalphabetic"],
    capture_output=True,
    text=True
)

if compile_result.returncode != 0:
    print("Compilation error:")
    print(compile_result.stderr)
else:
    # Execute the program
    result = subprocess.run(
        ["./monoalphabetic"],
        capture_output=True,
        text=True
    )
    print(result.stdout)
    # Save output for the lab record
    with open("monoalphabetic_output.txt", "w") as f:
        f.write(result.stdout)

    print("C program saved as monoalphabetic.c")
    print("Output saved as monoalphabetic_output.txt")
```

OUTPUT:

```
Plaintext: HELLO
Substitution alphabet: QWERTYUIOPASDFGHJKLZXCVBNM
Ciphertext: ITSSG
```

RESULT

Thus, the C program for the monoalphabetic substitution cipher was implemented and executed successfully, encrypting the plaintext HELLO into the ciphertext ITSSG using a unique substitution alphabet.